

PADHAISHURU

GATE Preparation Platform

GATE ME

Mechanical Engineering

TREND-BASED MOCK PAPER

— 03 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	11 subjects

Difficulty Distribution

Easy: 23 questions (35%)
Moderate: 33 questions (51%)
Difficult: 9 questions (14%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE ME — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [-0.33]

[General Aptitude]

Area of circle radius r:

- (A) $2\pi r$
- (B) πr^2
- (C) πr
- (D) $4\pi r^2$

Q2. 1M [-0.33]

[General Aptitude]

Antonym of 'prolific':

- (A) Scarce
- (B) Abundant
- (C) Creative
- (D) Unusual

Q3. 1M [no -ve]

[Engineering Mathematics]

Derivative of $\sin(x)$:

Enter your answer here

Q4. 1M [-0.33]

[Strength of Materials]

Stress = Force /:

- (A) Length
- (B) Area
- (C) Volume
- (D) Mass

Q5. 1M [-0.33]

[Strength of Materials]

Section modulus Z =:

- (A) I/y
- (B) I/G
- (C) M/Z
- (D) $\frac{1}{2} \pi r^3$

Q6. 1M [no -ve]

[Fluid Mechanics]

Laminar flow has:

Enter your answer here

Q7. 1M [no -ve]

[Fluid Mechanics]

Bernoulli applies to:

Enter your answer here

Q8.

1M [no -ve]

[Manufacturing Processes]

MIG welding uses:

Enter your answer here

Q9.

1M [-0.33]

[Manufacturing Processes]

Sand casting uses:

- (A) Permanent mold
- (B) Sand mold
- (C) Die
- (D) Investment

Q10.

1M [no -ve]

[Thermodynamics]

First law based on:

Enter your answer here

Q11.

1M [-0.33]

[Thermodynamics]

Entropy change for reversible process:

- (A) Positive
- (B) Negative
- (C) Zero
- (D) Infinity

Q12.

1M [-0.33]

*[Heat Transfer]*Stefan-Boltzmann: $q = ?$

- (A) $5 \times 10^{-8} \text{ W/m}^2\text{K}^4$
- (B) $5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
- (C) $5.67 \times 10^{-10} \text{ W/m}^2\text{K}^4$
- (D) $5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$

Q13.

1M [-0.33]

*[Heat Transfer]*Fourier's law $q =$:

- (A) $-k \frac{dT}{dx}$
- (B) $k \frac{dT}{dx}$
- (C) $k T$
- (D) $-k T$

Q14.

1M [-0.33]

[Theory of Machines]

Cam converts:

- (A) Rotary to rotary
- (B) Rotary to reciprocating
- (C) Recip to recip
- (D) None

Q15.

1M [no -ve]

[Theory of Machines]

Gear ratio is:

*Enter your answer here***Q16.**

1M [no -ve]

[Engineering Mechanics]

Equilibrium requires:

*Enter your answer here***Q17.**

1M [-0.33]

[Engineering Mechanics] $F = ma$ is:

- (A) Newton 1st
- (B) 2nd
- (C) 3rd
- (D) None

Q18.

1M [-0.33]

[Machine Design]

Key prevents:

- (A) Sliding
- (B) Rotation
- (C) Both
- (D) None

Q19.

1M [no -ve]

[Machine Design]

Shaft designed for:

*Enter your answer here***Q20.**

1M [no -ve]

[Thermodynamics]

Entropy change for reversible process:

*Enter your answer here***Q21.**

1M [no -ve]

*[Heat Transfer]*Fourier's law $q =$:

*Enter your answer here***Q22.**

1M [-0.33]

[Manufacturing Processes]

MIG welding uses:

- (A) Flux
- (B) Gas shield
- (C) Both
- (D) None

Q23.

1M [no -ve]

[Manufacturing Processes]

Sand casting uses:

*Enter your answer here***Q24.**

1M [-0.33]

*[Engineering Mathematics]*Derivative of $\sin(x)$:

- (A) $\cos(x)$
- (B) $-\sin(x)$
- (C) $-\cos(x)$
- (D) $\tan(x)$

Q25.

1M [-0.33]

[Machine Design]

Shaft designed for:

- (A) Torsion
- (B) Bending
- (C) Combined
- (D) Axial

Q26.

1M [-0.33]

[Machine Design]

Shaft designed for:

- (A) Torsion
- (B) Bending
- (C) Combined
- (D) Axial

Q27.

1M [-0.33]

[Engineering Mechanics]

Equilibrium requires:

- (A) $\sum \vec{F} = 0$
- (B) $\sum \vec{M} = 0$
- (C) Both
- (D) None

Q28.

1M [-0.33]

[Theory of Machines]

Gear ratio is:

- (A) Teeth ratio
- (B) Speed ratio
- (C) Torque ratio
- (D) Power ratio

Q29.

1M [no -ve]

[Machine Design]

Shaft designed for:

Enter your answer here

Q30.

1M [-0.33]

[Fluid Mechanics]

Bernoulli applies to:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) All

Q31.

2M [-0.66]

[Engineering Mechanics] $F = ma$ is:

- (A) Newton 1st
- (B) 2nd
- (C) 3rd
- (D) None

Q32.

2M [-0.66]

*[Heat Transfer]*Stefan-Boltzmann: $q = ?$

- (A) $<5 \times 10^{-8} \text{ W/m}^2\text{K}^4$
- (B) $<5 \times 10^{-5} \text{ W/m}^2\text{K}^4$
- (C) $<5 \times 10^{-10} \text{ W/m}^2\text{K}^4$
- (D) $<5 \times 10^{-12} \text{ W/m}^2\text{K}^4$

Q33.

2M [no -ve]

[IC Engine]

Otto cycle is:

Enter your answer here

Q34.

2M [-0.66]

*[Heat Transfer]*Stefan-Boltzmann: $q = ?$

- (A) $<5 \times 10^{-8} \text{ W/m}^2\text{K}^4$
- (B) $<5 \times 10^{-5} \text{ W/m}^2\text{K}^4$
- (C) $<5 \times 10^{-10} \text{ W/m}^2\text{K}^4$
- (D) $<5 \times 10^{-12} \text{ W/m}^2\text{K}^4$

Q35.

2M [-0.66]

[Thermodynamics]

First law based on:

- (A) Energy conservation
- (B) Entropy
- (C) Temperature
- (D) Pressure

Q36.

2M [-0.66]

[Manufacturing Processes]

MIG welding uses:

- (A) Flux
- (B) Gas shield
- (C) Both
- (D) None

Q37.

2M [-0.66]

*[Engineering Mathematics]*Derivative of $\sin(x)$:

- (A) $\cos(x)$
- (B) $-\sin(x)$
- (C) $-\cos(x)$
- (D) $\tan(x)$

Q38.

2M [-0.66]

[Thermodynamics]

Entropy change for reversible process:

- (A) Positive
- (B) Negative
- (C) Zero
- (D) Infinity

Q39.

2M [no -ve]

[Thermodynamics]

First law based on:

*Enter your answer here***Q40.**

2M [-0.66]

[Machine Design]

Key prevents:

- (A) Sliding
- (B) Rotation
- (C) Both
- (D) None

Q41.

2M [-0.66]

[Manufacturing Processes]

Sand casting uses:

- (A) Permanent mold
- (B) Sand mold
- (C) Die
- (D) Investment

Q42.

2M [no -ve]

[General Aptitude]

Area of circle radius r:

- (A) $2\pi r$
- (B) πr^2
- (C) πr
- (D) $4\pi r^2$

Q43.

2M [-0.66]

[Engineering Mechanics]

F = ma is:

- (A) Newton 1st
- (B) 2nd
- (C) 3rd
- (D) None

Q44.

2M [-0.66]

*[Heat Transfer]*Stefan-Boltzmann: $q = ?$

- (A) σT^4
- (B) σT^5
- (C) σT^3
- (D) σT^2

Q45.

2M [no -ve]

[General Aptitude]

Area of circle radius r:

Enter your answer here

Q46.

2M [-0.66]

*[Engineering Mathematics]*Derivative of $\sin(x)$:

- (A) $\cos(x)$
- (B) $-\sin(x)$
- (C) $-\cos(x)$
- (D) $\tan(x)$

Q47.

2M [-0.66]

[Engineering Mechanics]

F = ma is:

- (A) Newton 1st
- (B) 2nd
- (C) 3rd
- (D) None

Q48.

2M [-0.66]

[Fluid Mechanics]

Laminar flow has:

- (A) Low Re
- (B) High Re
- (C) $Re=1$
- (D) $Re=0$

Q49.

2M [-0.66]

*[Heat Transfer]*Fourier's law $q =$:

- (A) $-k \, dT/dx$
- (B) $k \, dT/dx$
- (C) $k \, T$
- (D) $-k \, T$

Q50.

2M [-0.66]

*[Heat Transfer]*Fourier's law $q =$:

- (A) $-k \, dT/dx$
- (B) $k \, dT/dx$
- (C) $k \, T$
- (D) $-k \, T$

Q51.

2M [-0.66]

*[Heat Transfer]*Stefan-Boltzmann: $q = ?$

- (A) σT^4
- (B) σT^3
- (C) σT^2
- (D) σT

Q52.

2M [no -ve]

*[Strength of Materials]*Section modulus $Z =$:

Enter your answer here

Q53.

2M [no -ve]

[IC Engine]

Otto cycle is:

Enter your answer here

Q54.

2M [-0.66]

*[Strength of Materials]*Section modulus $Z =$:

- (A) I/y
- (B) I/G
- (C) M/Z
- (D) σ/ϵ

Q55.

2M [no -ve]

[Fluid Mechanics]

Laminar flow has:

*Enter your answer here***Q56.**

2M [no -ve]

[Strength of Materials]

Stress = Force /:

*Enter your answer here***Q57.**

2M [-0.66]

*[Heat Transfer]*Fourier's law $q =$:

- (A) $-k \, dT/dx$
- (B) $k \, dT/dx$
- (C) $k \, T$
- (D) $-k \, T$

Q58.

2M [-0.66]

[Fluid Mechanics]

Laminar flow has:

- (A) Low Re
- (B) High Re
- (C) $Re=1$
- (D) $Re=0$

Q59.

2M [no -ve]

[Strength of Materials]

Stress = Force /:

*Enter your answer here***Q60.**

2M [-0.66]

*[Strength of Materials]*Section modulus $Z =$:

- (A) I/y
- (B) I/G
- (C) M/Z
- (D) $<7'0^\bullet$

Q61.

2M [no -ve]

[Manufacturing Processes]

MIG welding uses:

Enter your answer here

Q62.

2M [no -ve]

[Machine Design]

Shaft designed for:

*Enter your answer here***Q63.**

2M [no -ve]

[Manufacturing Processes]

Sand casting uses:

*Enter your answer here***Q64.**

2M [no -ve]

*[Heat Transfer]*Fourier's law $q =$:

*Enter your answer here***Q65.**

2M [-0.66]

[Manufacturing Processes]

Sand casting uses:

- (A) Permanent mold
- (B) Sand mold
- (C) Die
- (D) Investment